

# From CNNs to GANs for cross-modality medical image estimation

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## ABSTRACT

Cross-modality image estimation involves the generation of images of one medical imaging modality from that of another modality. Convolutional neural networks (CNNs) have been shown to be useful in image-to-image intensity projections, in addition to identifying, characterising and extracting image patterns. Generative adversarial networks (GANs) use CNNs as generators and estimated images are classified as true or false based on an additional discriminator network. CNNs and GANs within the image estimation framework may be considered more generally as deep learning approaches, since medical images tend to be large in size, leading to the need for large neural networks. Most research in the CNN/GAN image estimation literature has involved the use of MRI data with the other modality primarily being PET or CT. This review provides an overview of the use of CNNs and GANs for cross-modality medical image estimation. We outline recently proposed neural networks and detail the constructs employed for CNN and GAN image-to-image synthesis. Motivations behind cross-modality image estimation are outlined as well. GANs appear to provide better utility in cross-modality image estimation in comparison with CNNs, a finding drawn based on our analysis involving metrics comparing estimated and actual images. Our final remarks highlight key challenges faced by the cross-modality medical image estimation field, including how intensity projection can be constrained by registration (unpaired versus paired data), use of image patches, additional networks, and spatially sensitive loss functions.

## 1. Introduction

Medical imaging is the process of generating images of body parts and organs to help diagnose diseases and disorders, plan treatments and monitor disease progression. The different types of medical imaging devices provide information complementing clinical findings. Magnetic resonance imaging (MRI) provides information on soft tissues, computed tomography (CT) is mostly used to image high electron density tissues such as bone but can also provide a level of soft tissue contrast, and molecular imaging techniques, namely single photon emission computed tomography (SPECT) and positron emission tomography (PET), image specific biological functions using radiotracers. Multi-modality medical images are now integral to clinical decision making. However, acquisition of images using multiple modalities has distinct disadvantages because multiple scans increase cost and radiation dose and delay clinical workflow. Cross-modality image estimation has the potential to overcome specific, existing limitations.

Cross-modality image estimation involves the synthesis of medical images between two different imaging modalities. Benefits of cross-

modality image estimation from a patient perspective include fewer scans, less delay, and lower radiation dose and healthcare costs. The medical provider benefits also by having faster scanning turnaround and higher patient throughput, reduced staffing and reduced healthcare and maintenance costs, on top of lower radiotracer production runs. Registration mismatch across modalities is also eliminated with cross-modality image estimation, as images are synthesised to the same coordinate and orientation system. Whilst these are important advantages of cross-modality image estimation, careful consideration should be made towards the validity of image estimation for clinical use. This is because the physics governing image formation for the different modalities is fundamentally distinct. MRI has been of high interest in cross-modality image estimation, as it is a non-invasive imaging modality, i.e., no ionising radiation exposure to patients.

Multi-modality medical images came to feature in routine clinical practice around the early 1990s, and cross-modality image estimation methods have been of interest since about 2010. Image estimation methods fall into three categories – segmentation, atlas and machine learning based. Machine learning based image estimation has only been popular recently, as the tools continue to be developed. In the medical

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**Abbreviations used**

|                     |                                             |         |                                         |
|---------------------|---------------------------------------------|---------|-----------------------------------------|
| <sup>18</sup> F-FDG | Fluorodeoxyglucose (18F)                    | LSGAN   | Least Square GAN                        |
| CC                  | Correlation Coefficient                     | MAE     | Mean Absolute Error                     |
| CNN                 | Convolutional neural network                | MSE     | Mean Squared Error                      |
| CAE-GAN             | Conditional Auto-Encoder GAN                | NMSE    | Normalised Mean Squared Error           |
| cGAN                | Conditional GAN                             | N/A     | Not applicable                          |
| DSC                 | Dice Similarity Coefficient                 | ND      | Not Disclosed                           |
| DCNN                | Deep CNN                                    | NS      | Not Stated                              |
| DECNN               | Deep Embedding Convolutional Neural Network | PSNR    | Peak Signal-to-Noise Ratio              |
| FCN                 | Fully Convolutional Network                 | PCC     | Pearson Correlation Coefficient         |
| FCM                 | Fuzzy C-Means                               | ReLU    | Rectified Linear Unit                   |
| GPU                 | Graphics Processing Unit                    | ResNet  | Residual Network                        |
| GAN                 | Generative adversarial network              | SNR     | Signal-to-noise ratio                   |
| HECT                | High-energy CT                              | SSIM    | Structural Similarity Index Measure     |
| HU                  | Hounsfield Unit                             | UTE     | Ultrashort Echo-Time                    |
| LA-GAN              | Locality-adaptive GAN                       | UNIT    | Unsupervised Image-to-image Translation |
| LECT                | low-energy CT                               | VAE     | Variational Auto-encoder                |
|                     |                                             | ZeDD-CT | Zero-Echo-Time and Dixon Deep Pseudo-CT |

imaging field, supervised machine learning methods, particularly regression techniques, have been employed for image estimation. The convolutional neural network (CNN) within the deep learning framework has increasingly been used for multi-modality image estimation with mounting success. This review provides an overview of recent achievements in CNN-based image estimation, how the field has progressed towards generative adversarial networks (GANs) based on CNN constructs, and where data is available, how these methods benchmark against other image estimation techniques.

### 1.1. The need for cross-modality image estimation

According to methods reported in the literature, the application of CNNs/GANs for cross-modality image estimation fall into three primary groups. These include the synthesis of the CT and PET image from MRI, and the estimation of high resolution and high signal-to-noise ratio (SNR) PET images from low resolution and low SNR PET images using MRI as an anatomical high-resolution reference. Other combinations have also been considered, but to a much smaller extent and generally without a clear medical imaging justification.

#### 1.1.1. CT image synthesis from MRI

The superior soft tissue contrast and avoidance of radiation exposure has led to MRI replacing CT in many clinical applications. However, CT images still have to be acquired for radiotherapy treatment dose calculation and for PET attenuation correction for accurate image reconstruction [1]. The increased uptake of PET-MRI hybrid systems led to other investigations in which MRI was used to estimate PET attenuation correction maps normally derived from CT scans. Essentially, the relationship between the CT image and attenuation map is a transformation from Hounsfield units to a linear attenuation coefficient. The attenuation coefficient in high electron density tissues such as bone is large, while in soft tissue it is a small number. In CT bone appears hyperintense and in MRI hypointense or as a null signal, but soft tissue contrast around bone in MRI provides a mechanism by which bone patterns can be delineated.

#### 1.1.2. MRI synthesis from CT

CT scans are generally cheaper than MRI, and in some situations, MRI is not as widely available as a CT scan. Additionally, claustrophobia is more likely in an MRI in comparison to CT, due to longer scan times and longer scanner bores. Patients with non-MRI compliant implants and pacemakers are also excluded from having an MRI scan. These challenges have provided a motivation to estimate MRI from CT [2]. A

somewhat surprising application of MRI-CT synthesis and vice versa is image registration. It has been suggested that by first synthesising an image, registration of the synthetic image to the actual image can be performed more robustly [3]. The registration matrix obtained for the synthetic to actual image transform can then be applied to the other imaging modality directly.

#### 1.1.3. Estimating the PET image from MRI

It can be difficult to acquire imaging data in certain patient cohorts. Instances arise in the clinical setting where patients who have undergone MRI scans were not able to complete a PET scan. An example of such a case is Alzheimer's disease [4]. Here, MRI can be considered for estimating <sup>18</sup>F-FDG-PET images, as it is known that the <sup>18</sup>F-FDG tracer is taken up by the brain in general, resulting in PET soft tissue anatomical contrast. This provides the basis for MRI, a soft tissue contrast imaging modality, to be used for PET image estimation. It is important to note that to perform cross-modality image estimation, a level of information correspondence must exist between the imaging modalities. Hence, PET images of highly specific radiotracers with little or no anatomical image contrast are unlikely predictable from soft tissue contrast MRI scans. This is because the synthesised image is expected to contain the same anatomical features as the source image, however with a different pixel intensity distribution (i.e., non-linear intensity projection).

#### 1.1.4. Resolution and SNR boost in PET using MRI

PET images are inherently lower resolution than MRI scans due to the nature of the medical imaging modality. The 2–3 mm–1 mm mismatch in image resolution provides several disadvantages. These include anatomical and functional mismatch and blurring between the modalities, and reduced specificity of regions identified using PET, especially important in oncology and neurology [5]. The approach to bridge this gap has been to incorporate anatomical priors from MRI within a machine learning framework with the outcome of estimating PET images which match the MRI resolution.

Radiotracer dosimetry in PET is an important factor which influences image quality and SNR. It is desirable to reduce the PET dose given to a patient due to risks associated with radiation exposure [6]. However, low dose PET studies inherently lead to low SNR PET images. Whilst image SNR can be increased by raising the radiotracer dose, the preferred methods in low-dose PET studies are post-processing approaches for noise reduction. As such, machine learning approaches which take noisy images and MRI anatomical images as input and produce low noise PET images as an output have been of interest.

## 2. Methodology

### 2.1. Datasets

Cross-modality data is usually acquired with different image specifications (i.e., matrix size, resolution, number of slices, acquisition time, and contrast mechanism). Furthermore, images within the same or other modality differ in orientations (due to subject position, sensors, and type of acquisition in different scanners). It is also worth pointing out that image resolution differs across the modalities. Typically, CT, and MRI produce images with resolution greater than PET.

Some image pre-processing is essential for the machine to learn the cross-modality intensity projection robustly. The first pre-processing step is broad view alignment, which ensures that FOV of the input and target images used for training have the same axial, sagittal or coronal orientation, with consistent foreground-background ratios.

The common approach is often to register and resample the input images to the target images. Sometimes, an arbitrary resolution between the two modalities maybe chosen, requiring the resampling of images from both modalities. As such, the machine is assumed to have to only learn the intensity projection between the modalities and not any spatial mismatch between anatomical regions in images. When the registration is not perfect, or images are assumed to be aligned broadly, anatomical priors need to be considered. These can be in the form of a binary mask capturing the spatial misalignment between images, or in the form of image landmarks. Here, images are considered as unpaired. Paired data means input and output images have been aligned via registration. The final step is to crop the image background to minimise computational complexity by ensuring the intensities in the image mostly cover the FOV. Image-to-image intensity projection is then established using a machine learning approach.

### 2.2. Neural networks

#### 2.2.1. Convolutional Neural Network (CNN)

A CNN is an artificial neural network (ANN) consisting of an input layer, one or more hidden layers, and an output layer. By definition, a CNN has a minimum of one convolutional layer (hidden) incorporated into the network. This layer performs a convolution operation on the input. The output of the convolution layer is a feature map generally of a smaller size and becomes the input to the next layer. Deep learning

extends the CNN framework to large datasets with highly complex information, which necessitates the use of large neural networks [7].

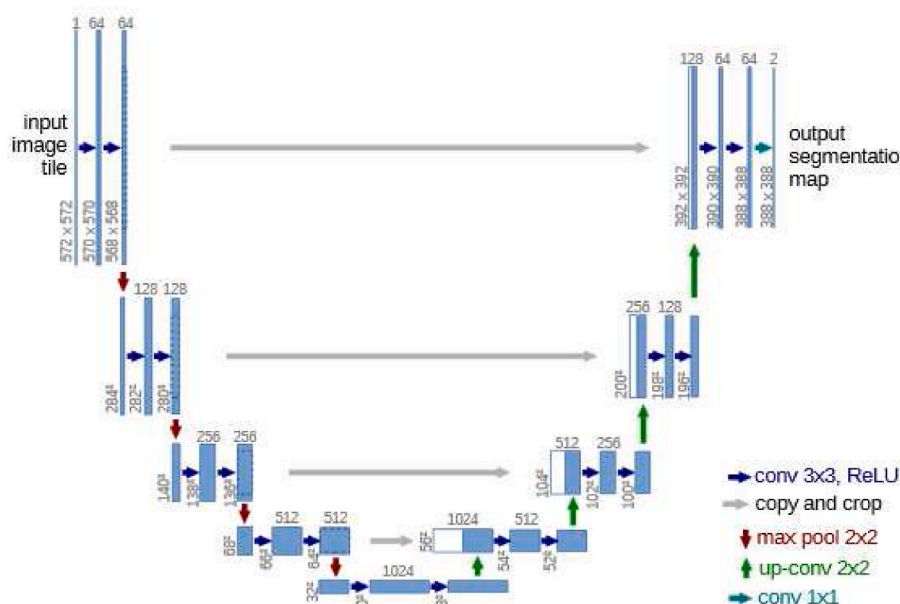
CNNs typically start with a convolutional layer followed by activation functions. The subsequent layer might be a convolutional layer or a pooling layer to down-sample image features. Depending on the CNN architecture, up-sampling layers, such as un-pooling and transposed convolutional layers, may additionally be introduced into the network to enlarge feature map size. At the end of the network at least one fully connected layer is implemented to output an image with specific features. Within a CNN encoder-decoder framework (see U-Net shown in Fig. 1), the encoder part has the action of dimensionality reduction into compressed features, and the decoder takes the compressed features and expands them into a target image.

**2.2.1.1. Down-sampling layers.** Convolutional layers mimic the convolution mathematical operation by applying a kernel filter to the input. Essentially, the convolution layer connection weights define the kernel, see U-Net and SegNet encoders in Figs. 1 and 2. The output of the layer is a new feature map determined by the kernel operation. In imaging applications, the filters are typically 2D or 3D matrices arranged based on the network dimensions.

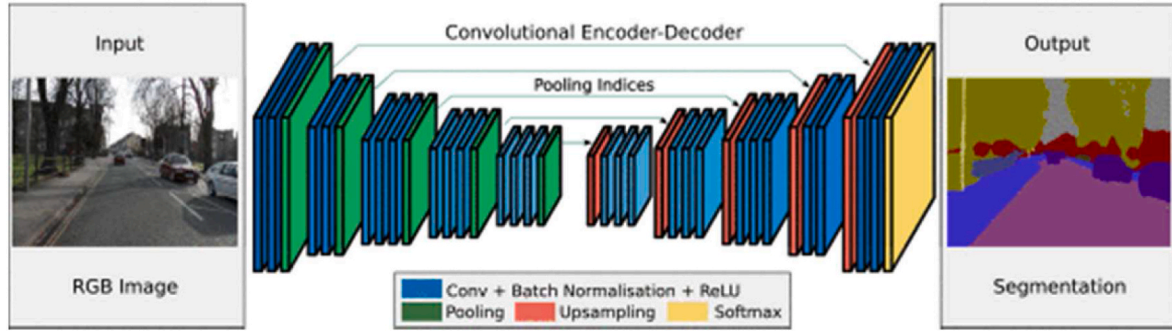
Pooling layers perform a non-trainable operation and they are used to reduce the complexity in the input data by ‘pooling’ values (see U-Net and SegNet encoder in Figs. 1 and 2). The most common pooling operation is max pool, which returns the maximum value in a select pooling window. The reduction in data size is determined by the size of the pooling window.

**2.2.1.2. Up-sampling layers.** Within the CNN architecture, up-sampling layers are used to increase output dimensionality. The un-pooling layer (see decoder for SegNet in Fig. 2), which is a non-trainable operation, projects a single value into a window via zero padding. The transposed convolutional layer will distribute a value into a window as defined by the kernel weights (see decoder for U-Net in Fig. 1).

**2.2.1.3. Fully connected layers.** The fully connected network uses fully connected layers at any point within a network. This layer provides a mechanism for making and retaining distal relationships in images. In imaging, such a layer allows connections to be retained between features, potentially defining anatomical correlates with distinct image intensities, at a distance from each other. Fully connected layers also



**Fig. 1.** The U-Net architecture is depicted. Blue boxes are multi-channel feature maps where information about the number of channels is provided on top of each box and the size of the feature maps are on their sides. The first half of the network is responsible for compressing (encoder) the feature maps through convolutional and pooling operations. The second half is responsible for decompressing (decoder) the feature maps and resize them to the original size by concatenating the corresponding feature maps followed by convolutional operations. The U-Net auto-encoder diagram has been adapted from Ref. [8]. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)



**Fig. 2.** SegNet architecture consists of encoder and decoder networks, here each with 13 convolutional layers. Up-sampling is performed by pooling indices obtained during encoding, and it is therefore transferred from the corresponding locations in the encoder network. This specific encoder-decoder network was described in Ref. [9].

provide a way of retaining spatial relationships in the data when 2D or 3D images are converted to 1D vectors of image intensities for the CNN in input.

**2.2.1.4. Activation functions.** Activation functions are predefined mathematical operations used to impose linear or nonlinear operations within the neural network. Output of the convolution operation is processed through an activation function before passed to the next layer. Different activation functions may be used after each layer. The choice of activation function in the network is very important and network performance is impacted by the choice. A popular activation function for imaging is the rectified linear unit (ReLU), which maps a negative value intensity 0 (i.e., 0 for  $x < 0$ ) and the positive value remains unchanged (i.e.,  $x$  for  $x \geq 0$ ).

**2.2.1.5. From image metrics to loss functions.** Three different metrics are commonly used in evaluating image estimation quality. These include mean absolute error (MAE), mean squared error (MSE) and peak signal-to-noise ratio (PSNR):

$$MAE = \frac{1}{N} \sum_{n=1}^N |I(n) - I_e(n)|, \quad (1)$$

$$MSE = \frac{1}{N} \sum_{n=1}^N (I(n) - I_e(n))^2, \quad (2)$$

$$PSNR = 10 \log_{10} \frac{MAX^2}{MSE}, \quad (3)$$

where  $N$  is the total number of pixels in an image,  $I(n)$  is an image intensity in the original image at location  $n$  and  $I_e(n)$  is the corresponding estimated value, and  $MAX$  is the maximum pixel value in the image data. The MAE is essentially an L1-norm loss, the MSE is the L2-norm loss, and PSNR is akin to  $L_{\infty}$ -norm loss. Other types of loss functions have been developed as well, and these will be highlighted in their own contexts.

The structural similarity index (SSIM), dice similarity coefficient (DSC) and Pearson correlation coefficient (PCC) have been used as well to measure how well the estimated image pixel intensities correlate with those in the real image. Image quality estimation improves with decreasing MAE and MSE and increasing DSC and PCC.

**2.2.1.6. CNN optimisation.** The CNN architecture consists of layers, connections between layers and weights within layers. The number, type and size of layers are defined by the user. The layer specific weights are determined through a process of CNN training and testing. This requires matching input and output data. As such, application specific training and testing data are requirements for machine learning. Ideally, the prediction produced by the CNN should match the expected output, and the difference is measured using a loss function. Different types and

weighted combination of loss functions could be used. The loss function figure-of-merit is minimised through the training cycles, known as epochs. The CNN is evaluated using a testing dataset made up of images not used during training. If the loss function figure-of-merit is comparable between training and testing, then the CNN architecture and layer specific weights are said to be optimised. Otherwise, either underfitting (not enough CNN architecture complexity to capture features in the data) or overfitting (too much CNN architecture complexity) may have occurred.

## 2.2.2. Generative Adversarial Network (GAN)

GAN is an extension of the CNN framework where two networks are trained simultaneously, a so-called generator and discriminator. The generator network is akin to those used in CNNs, a primary example of which is the U-Net. Essentially, the generator performs the image estimation, and the discriminator network check whether it is a real image, or not. The two networks therefore compete against each other, and the discriminator loss is minimised during training. Training halts when the discriminator network cannot distinguish whether the image produced by the generator is fake based on some predefined threshold.

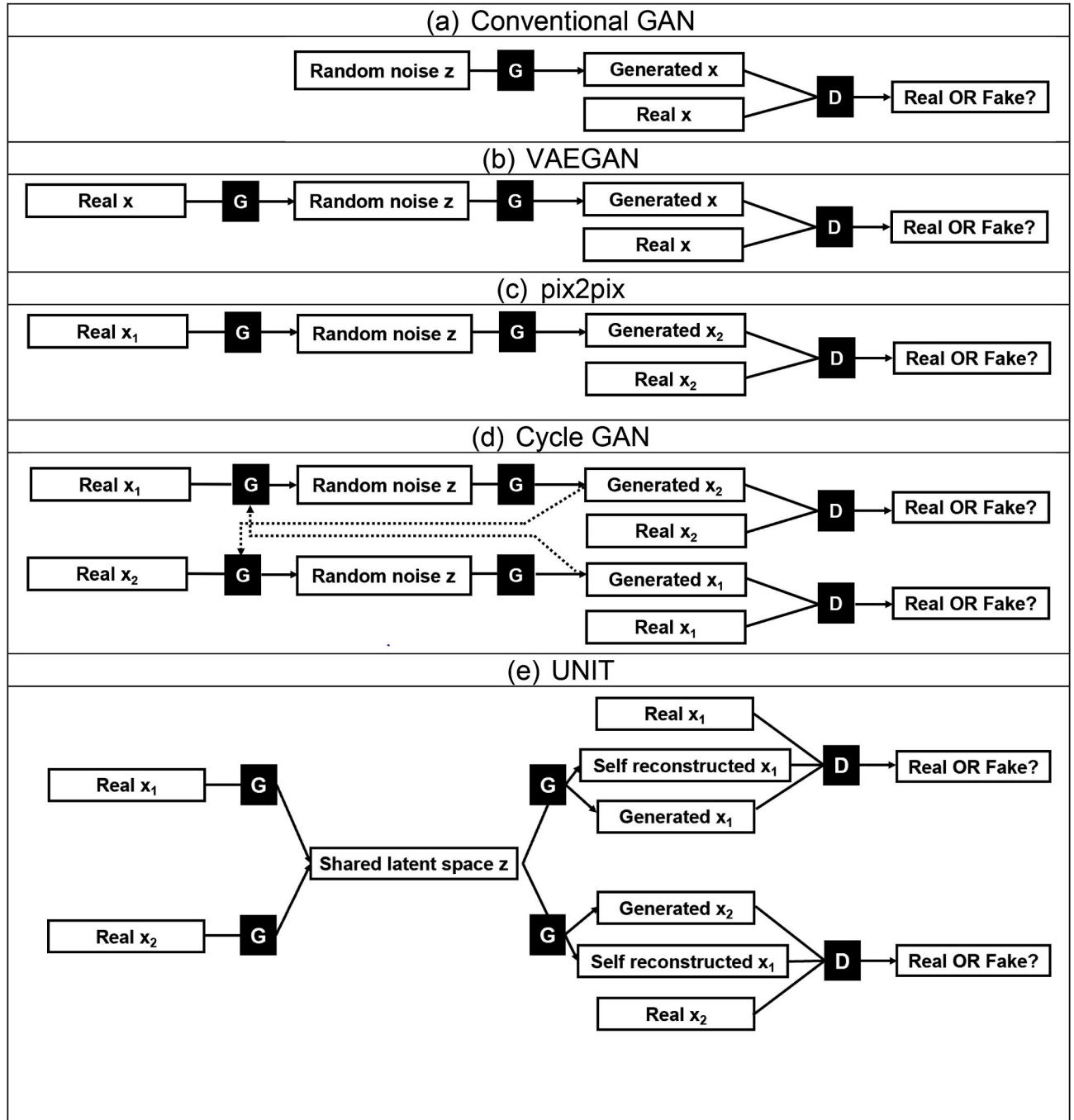
A challenge of GANs is to achieve a balance between the generator and the discriminator. One may dominate the other, and most likely the generator based on previous experience [10]. Here, the discriminator cannot distinguish fake images produced by the generator from real images. Conversely, if the discriminator dominates, then the generated images are easily distinguishable from real images, and the discriminator training rate stalls, and consequently training of the generator ends. This is a common problem associated with estimating high resolution images, as high spatial variability in images challenge the network [10]. An additional consideration is mode collapse, defined as the generator producing limited sample variety, occurring when the network is insufficiently large for the input data.

**2.2.2.1. GAN architectures for cross-modality image estimation.** The GAN frameworks provided in Table 1 can, in principle, all be used to perform intensity projection between images. The conditional GAN (cGAN) additionally constrains the output by placing conditions on the generator or the discriminator [11]. The use of conditions helps constrain the image estimates towards the desired outcome [12]. The cGAN layouts provided in Table 1 have been investigated in cross-modality image estimation, and a brief description is provided here:

- i) **VAEGAN** - A variational auto-encoder (VAE) network paired with a GAN in a way which allows sharing of parameters across the two networks within a joint training framework [13,14].
- ii) **pix2pix** - A CNN, often the auto-encoder type, is used as the generator and the discriminator checks how the estimated image resembles the actual image [10,15]. A loss function specific to the

**Table 1**

Shown are the conventional GAN and different types of conditional GANs used in cross-modality medical image estimation.



pix2pix architecture is used, which generalises across different applications.

iii) CycleGAN – This type of GAN allows network training when accurate spatial correspondence between the input and target images is lacking. Here, the standard adversarial loss does not suffice, leading to the use of the cycle consistency loss. CycleGANs have been applied on unpaired [16] and paired data [17], with the latter study suggesting performance improvements with unpaired data.

iv) UNIT - An Unsupervised Image-to-image Translation (UNIT) neural network consisting of a coupled pair of VAEGANs [18,19]. Each VAEGAN in a generator network applies to one imaging modality and the latent space is shared between generators [20,21].

### 3. Cross-modality image estimations

Imaging data can be in the form of image slices (i.e., 2D) or 3D volumes. Whilst people have been developing neural networks which take 2D and 3D medical imaging data, the limitation of these networks is the large number of trainable network nodes (note, trainable nodes scale with input-output image size). This also leads to the need for large training datasets, which are often hard to come by for patient cohorts. Because of this, there have been studies which subsample images in the form of patches, to be able to work with smaller neural networks and to increase the available training data, since many patches can be extracted from an image. Whilst today GANs appear to have superseded CNNs for

cross-modality image estimation, we provide a review of the CNN literature to have an appreciation from a historical perspective and to understand how CNNs have shaped the GAN literature. [Tables 4 and 5](#) summarise CNN and GAN architectures involved in this review.

### 3.1. CNN implementations using patches

Li et al. [40] trained a 3D patch-based CNN to estimate PET from MRI, and applied it to recover missing data in PET. The CNN here appeared to work by creating an optimised network which learnt the intensity projection between the two modalities within the human brain based on anatomical intensity variations in the MRI data. To estimate pelvic CT from corresponding MRI, Nie et al. [84] proposed a patch-based 3D fully convolutional CNN building on the Caffe architecture [85]. The inherent use of fully convolutional networks (FCN) led to improved preservation of spatial information. Subsequently, Roy et al. [25] synthesised CT from MRI using two-channel inception blocks via GoogleNet [86]. The use of ultra-short echo time (UTE) MRI instead of T1-weighted MRI provided better sensitivity to bone, the result of which was PSNR improvement in the estimated CT images. These results suggest that zero intensity regions projected to non-zero values challenge these image synthesis methods.

Xiang et al. [6] investigated low dose PET image enhancement with the aid of MRI applied in the auto-context CNN network. The implementation involved the concatenation of low-dose PET and MRI patch features after every four layers in the neural network. The auto-context CNN outperformed the conventional approach in terms of PSNR (24.76 vs 23.98) and NMSE (0.0206 vs 0.0247). The use of T1-weighted MRI data outperformed the case when MRI data was not included (PSNR: 23.85 vs 23.11; NMSE: 0.0254 vs 0.0299). Example images are shown in [Fig. 3](#), and the result suggests anatomical priors from MRI improve the quality of the intensity projection. Xiang et al. later proposed a 3D patch-based deep embedding convolutional neural network (DECNN) resembling the auto-context model, with the purpose of synthesising CT from MRI in the brain and prostate regions [22]. Essentially, DECNN combines the embedding blocks to allow the use of single neural network. DECNN outperformed Atlas-based, random forest (a non-CNN learning method) and previously proposed CNN and FCN implementations, as can be deduced from the MAE and PSNR results in [Table 2](#).

Song et al. [5] proposed a multi-channel deep learning method using both shallow and deep networks. In addition to low-resolution PET and high-resolution T1-weighted MRI, the radial and axial spatial location of the patch was provided as input. They found a significant improvement in contrast-to-noise ratio, PSNR and SSIM, confirming results of others on the use of MRI anatomical priors. Patch location information was found to provide limited benefits, which may be due to spatial information having already been encoding in the MRI. The use of deep or shallow CNN led to additional improvements in PSNR and SSIM. However, qualitative differences between the shallow and deep CNN estimated images were hard to discern when MRI formed part of the input. This suggests that smaller networks can be used for intensity projection across imaging modalities when anatomical information is used to constrain the neural network.

**Table 2**

Provided are the mean MAE and mean PSNR with standard deviations for the CT synthesised results using state-of-the-art methods. Reproduced from work by Ref. [22].

| Remark    | Atlas           | RF             | CNN            | FCN            | DECNN          | Atlas         | RF            | CNN           | FCN           | DECNN         |
|-----------|-----------------|----------------|----------------|----------------|----------------|---------------|---------------|---------------|---------------|---------------|
|           | Brain           | Brain          | Brain          | Brain          | Brain          | Prostate      | Prostate      | Prostate      | Prostate      | Prostate      |
| Mean MAE  | 169.5<br>± 35.7 | 99.9 ±<br>14.2 | 93.6 ±<br>11.2 | 88.9 ±<br>10.6 | 85.4 ±<br>9.24 | 64.6 ±<br>6.6 | 48.1 ±<br>4.6 | 45.3 ±<br>4.2 | 42.4 ±<br>5.1 | 42.5 ±<br>3.1 |
| Mean PSNR | 20.9 ±<br>1.6   | 26.3 ±<br>1.4  | 26.7 ±<br>1.0  | 27.1 ±<br>1.3  | 27.3 ±<br>1.1  | 29.1 ±<br>2.0 | 32.1 ±<br>0.9 | 33.1 ±<br>0.9 | 33.4 ±<br>1.1 | 33.5 ±<br>0.8 |

**Table 3**

MAE, MSE and PCC metrics for CT estimation from MRI using the DCNN architecture compared with the Atlas-based result. This DCNN outperformance comparison has been reproduced from Ref. [23].

| Quality metric | DCNN  | Atlas-based |
|----------------|-------|-------------|
| MAE            | 84.8  | 94.5        |
| MSE            | 188.6 | 198.3       |
| PCC            | 0.906 | 0.896       |

### 3.2. CNN implementations on slices and volumes

Han [23] estimated brain CT from T1-weighted MRI using a 2D deep CNN (DCNN). A previously described U-Net architecture [8] was modified by removing the 3-fully connected layers, which allowed 90% reduction in the number of trainable parameters, and faster CNN training was achieved. [Table 3](#) summarises the outperformance of the DCNN over the benchmark Atlas-based method. The fully connected layers are often used to retain spatial relationships between image intensities. The findings here suggest that the spatial relationship can be relaxed with the use of registered images. Gupta et al. [34] showed that the Han's model can be adopted to head images.

Dinka et al. [31] used a 2.5D network to estimate CT from T2-weighted MRI data in the three primary field-of-view orientations. The 2.5D implementation was chosen to overcome potential discontinuities across slices in the estimated images. The dilation of convolutional filters was also used to replace the down-sampling layers (i.e., pooling) and improve estimation accuracy. In this framework dental artifacts in CT images, which are problematic for radiotherapy treatment planning, were circumvented in CT images synthesised from MRI. This observation can be attributed to having low sensitivity to dental artifacts in T2-weighted MRI. They also stated that radiotherapy treatment dose calculations based on the synthetic CT images were comparable to those using the actual CT images. Concurrently, Liu and Qi [41] adopted a U-Net for the purpose of improving PET image signal-to-noise ratio. A DCNN model using three U-Nets was proposed, whereby the PET and MRI data were fed to independent 1-U-Nets, before the output was concatenated into another U-Net, resulting in the 3-U-Net architecture. Their study concludes that the use of the MRI 1-U-Net in the 3-U-Net structure improves MSE (31.3% and 34.0% versus 31.3%), in conjunction with an image contrast grain from 1.4 to 2.7.

Spadea et al. [35] investigated the use of DCNN for CT synthesis for the purpose of MRI-based proton therapy to replace CT-based treatment planning. In the machine learning model they considered single plane versus multiplane networks with different field-of-view orientations, and showed the use of multiplanes leads to significantly better image estimations. Wang et al. [32] estimated CT from MRI in nasopharyngeal carcinoma using the U-Net architecture. Their radiotherapy treatment planning findings were consistent with those by others, and example images are provided in [Fig. 4](#).

Arabi et al. [26] estimated the CT image from MRI in the pelvic region, and concurrently, Chen et al. [27,28] used a similar 2D U-Net architecture for estimating CT from MRI for the prostate region.

**Table 4**

CNN architecture summary and parameters required for training, + represents MSE, \* represents MAE, “1” and “2” represents 1-norm (L1) and 2-norm (L2) loss function. N/A: Not Applicable, NS: Not Stated, ND: Not disclosed by author.

| Author-year | Image estimation                                      | Architecture                                                      | Remarks         | Sample size (Training-Testing) | Patch size                                                    | Number of patches (from each volume) | Number of Convolutional layers                       | Pooling layers Y/ N? | Error (HU) Mean $\pm$ standard deviation                                                                                                                                           |                                             |
|-------------|-------------------------------------------------------|-------------------------------------------------------------------|-----------------|--------------------------------|---------------------------------------------------------------|--------------------------------------|------------------------------------------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| [24]        | T2-weighted MR: sCT                                   | 3D FCN                                                            | Pelvic          | 22                             | MRI: $32 \times 32 \times 16$<br>CT: $24 \times 24 \times 12$ | 6000                                 | 2                                                    | N                    | $42.4 \pm 5.1^{*1}$                                                                                                                                                                |                                             |
| [25]        | (UTE) dual-echo MR: sCT                               | 3D CNN (Inception blocks adopted)                                 | Brain           | 6                              | $25 \times 25 \times 5$                                       | 500,000                              | 6                                                    | Y                    | PSNR:21.92 <sub>+2</sub>                                                                                                                                                           |                                             |
| [23]        | T1-Weighted MR: sCT                                   | 2D DCNN, (U-Net adopted)                                          | Brain           | 18                             | N/A                                                           | N/A                                  | 27                                                   | Y                    | $84.8 \pm 17.3^{*1}$                                                                                                                                                               |                                             |
| [18,19]     | MR: CT                                                | 2D Auto-encoder network                                           | Brain           | 30–10                          | N/A                                                           | N/A                                  | 26                                                   | Y                    | Multiclass cross-entropy loss                                                                                                                                                      |                                             |
| [26]        | T2-Weighted MR: sCT                                   | 2D DCNN (U-Net adopted)                                           | Pelvic          | 39                             | N/A                                                           | N/A                                  | 27                                                   | NS                   | $18.4 \pm 6.6^{*1}$ (bladder)<br>$78.3 \pm 69.2^{*1}$ (rectum)<br>$32.7 \pm 7.9^{*1}$ (body)<br>$119.9 \pm 22.6^{*1}$ (bone)                                                       |                                             |
| [27,28]     | T2-Weighted MR: sCT                                   | 2D DCNN (U-Net adopted)                                           | Pelvic          | 36–15                          | N/A                                                           | N/A                                  | 23                                                   | Y                    | $29.96 \pm 4.87^{*1}$ (within body)                                                                                                                                                |                                             |
| [1]         | ZTE MR + Dixon (fat and water) MR: sCT                | 3D DCNN (U-Net adopted)                                           | Pelvic          | 10–16                          | $32 \times 32 \times 16$                                      | 60,000                               | 13                                                   | NS                   | 3D:<br>$37.6 \pm 5.1^{*1}$                                                                                                                                                         |                                             |
| [29]        | T1-Weighted MR: sCT                                   | 2.5D dilated CNN                                                  | Brain           | 52                             | N/A                                                           | N/A                                  | 10                                                   | N                    | $67 \pm 11^{*1}$                                                                                                                                                                   |                                             |
| [22]        | T1-Weighted MR: sCT                                   | 3D DECNN (Embedding blocks)                                       | Brain<br>Pelvic | 16<br>22                       | $64 \times 64 \times 3$                                       | Brain: 600,000<br>Prostate:100,000   | 3 K + 5 (K indicates the number of embedding blocks) | 11<br>17             | N                                                                                                                                                                                  | $85.4 \pm 9.24^{*1}$<br>$33.5 \pm 0.8^{*1}$ |
| [30]        | T1-Weighted MR: sCT                                   | 2D & 3D CNN (U-Net adopted)                                       | Pelvic          | 20                             | N/A                                                           | N/A                                  | 27                                                   | Y                    | 2D:<br>$40.5 \pm 5.4^{*1}$                                                                                                                                                         |                                             |
| [31]        | T2-Weighted MR: sCT                                   | 3D (U-Net adopted)                                                | Head and Neck   | 22                             | $48 \times 48 \times 48$                                      | 8                                    | 14                                                   | NS                   | $75 \pm 9^{*1}$                                                                                                                                                                    |                                             |
| [32]        | T2-Weighted MR: sCT                                   | 2D (U-Net adopted)                                                | Head and Neck   | 23–10                          | N/A                                                           | N/A                                  | 23                                                   | Y                    | $97 \pm 13^{*1}$ (soft tissue),<br>$131 \pm 24^{*1}$ (overall region), $357 \pm 44^{*1}$ (bone)<br>$^{*1}$                                                                         |                                             |
| [33]        | Dixon -Vibe MR: sCT                                   | 2D (U-Net adopted)                                                | Pelvic          | 15–4                           | N/A                                                           | N/A                                  | 23                                                   | Y                    |                                                                                                                                                                                    |                                             |
| [34]        | T1-Weighted MR: sCT                                   | 2D DCNN, (U-Net adopted)                                          | Brain           | 47–13                          | N/A                                                           | N/A                                  | NS                                                   | NS                   | $17.6 \pm 3.4^{*1}$ (soft tissue)<br>$^{*1}$ & +2                                                                                                                                  |                                             |
| [35]        | T1-Weighted MR: sCT                                   | 2D DCNN, (U-Net adopted)                                          | Brain           | 15                             | N/A                                                           | N/A                                  | 27                                                   | Y                    | $54 \pm 7^{*1}$ (within the head)                                                                                                                                                  |                                             |
| [36]        | T1-weighted multi-echo gradient-echo acquisition: sCT | 4D (3 spatial dimensions and 1 channel dimension) (U-Net adopted) | Pelvic          | 17 canines, 23 humans          | $C \times 24 \times 24$ (“c” is number of channels)           | NS                                   | NS                                                   | Y                    | $33-40^{*1}$ (human)<br>$35-47^{*1}$ (canines)                                                                                                                                     |                                             |
| [37]        | T1-Weighted MR Dixon: sCT                             | 2.5D (U-Net adopted)                                              | Abdominal       | 46                             | N/A                                                           | N/A                                  | 23                                                   | Y                    | $24.10^{*1}$ (liver),<br>$28.62^{*1}$ (spleen),<br>$47.05^{*1}$ (kidneys),<br>$29.79^{*1}$ (spinal cord),<br>$105.68^{*1}$ (lungs),<br>$110.09^{*1}$ (vertebral bodies)<br>$^{*1}$ |                                             |
| [38]        | T2-Weighted MR: sCT                                   | 2D eCNN                                                           | Pelvic          | 12                             | N/A                                                           | N/A                                  | 52                                                   | Y                    | $30.0 \pm 10.4^{*1}$                                                                                                                                                               |                                             |
| [39]        | T1 & T2-Weighted MR: sCT                              | 4D (3 spatial dimensions and 1 channel dimension) (U-Net adopted) | Lumbar Spine    | 3                              | ND                                                            | ND                                   | ND                                                   | ND                   | ND                                                                                                                                                                                 |                                             |
| [40]        | T1-weighted MR: PET                                   | 3D CNN                                                            | Brain           | 199                            | $3 \times 3 \times 3$                                         | 50,000                               | 2                                                    | N                    | NS                                                                                                                                                                                 |                                             |
| [6]         | (Low-dose-PET + T1-Weighted                           | 2D CNN                                                            | Brain           | 16                             | Step 1:<br>$27 \times 27$<br>Step 2:                          | 30,000                               | 4                                                    | N                    | $0.0206^{+2}$                                                                                                                                                                      |                                             |

(continued on next page)

Table 4 (continued)

| Author-year | Image estimation                                                        | Architecture             | Remarks | Sample size (Training-Testing) | Patch size                                  | Number of patches (from each volume) | Number of Convolutional layers                      | Pooling layers Y/N? | Error (HU) Mean $\pm$ standard deviation |
|-------------|-------------------------------------------------------------------------|--------------------------|---------|--------------------------------|---------------------------------------------|--------------------------------------|-----------------------------------------------------|---------------------|------------------------------------------|
|             | MR): standard-dose-PET                                                  |                          |         |                                | $19 \times 19$<br>Step 3:<br>$11 \times 11$ |                                      |                                                     |                     |                                          |
| [27,28]     | (Low dose 18F-florbetaben PET + T1-W, T2-W & T2-FLAIR MR): high SNR PET | 2D DCNN, (U-Net adopted) | Brain   | 32–8                           | N/A                                         | N/A                                  | 22                                                  | Y                   | $0.143725^{+1}$                          |
| [41]        | (Low SNR PET + T1-Weighted ME): high SNR PET                            | 3U-Net (U-Net adopted)   | Brain   | 20                             | N/A                                         | N/A                                  | 26                                                  | Y                   | $+2$                                     |
| [5]         | (Low resolution PET + T1-Weighted MR): super-resolution PET             | 2D DCNN                  | Brain   | 15                             | NS                                          | NS                                   | Shallow CNN: 3<br>Deep CNN: 12<br>Very deep CNN: 20 | N                   | $*^1$ NS                                 |
| [42]        | (Low count PET + T1-Weighted MR): full count PET                        | 3D CNN                   | Brain   | 20                             | N/A                                         | N/A                                  | 3                                                   | N                   | $+2$                                     |

According to both studies, the body and soft tissue MAE were smaller using the CNN than that produced by the benchmark atlas-based method, and in bony structures the error was larger. In view of this shortcoming, Leynes et al. [1] used a 3D three-channel U-Net with the inputs including ZTE (i.e., zero echo time images to have some sensitivity to bone in the MRI) images along with Dixon fat and water images. The three-channel inputs outperformed single Dixon-CT input for PET attenuation correction, suggesting the additional use of ZTE MRI data provided better bone estimation, which becomes useful for producing attenuation correction maps from MRI otherwise obtained using CT.

Fu et al. [30] also focused on the pelvic region and estimated CT both using 2D and 3D data. Han's model was modified to allow for more rapid convergence during training. The modifications included replacing the un-pooling layers with fractionally-strided convolutional layers (like transpose convolution) to produce dense, high-resolution feature maps, instead of sparse, high-resolution ones normally achieved via un-pooling layers. Additionally, they implemented the residual shortcuts proposed for the ResNet architecture [87], leading to encoder feature maps (before down-sampling) which correspond with up-sampled feature maps (decoder) in the residual frame. Irrespective of the approach, images could be estimated using a trained network in 5.5s. The 3D network produced smaller MAEs in general (37.6 versus 40.5), with higher bone position accuracy ( $<0.6$  mm error), when compared to the 2D network. Their synthesised images are provided in Fig. 5 with image metrics in Fig. 6.

### 3.3. GAN implementations

As for CNNs, numerous GAN implementations have been investigated focusing on head, neck, pelvic, and cardiac regions with similar motivations justifying the use of image synthesis. Each advance aimed to overcome an existing image estimation limitation. Lei et al. [51] estimated CT from MRI in both head and pelvic regions using a dense CycleGAN in which the dense blocks were replaced with residual blocks in the generator, leading to improved information in estimated images. The use of a distance loss function in addition to a gradient difference loss function was able to compensate for blur and misclassification of image information. Notably, these loss functions both act on the spatial variations in intensities, providing a mechanism for spatially constraining the intensity projection across the modalities. Yang et al.'s [49] structure constrained CycleGAN involved the addition of a structural-consistency loss in addition to adversarial and

cycle-consistency losses to improve the quality of image estimation from MRI to CT in the head and neck regions. Instead of training the network using random image patches, they also chose patches to be oriented the same with approximately same locations. This training choice helped the discriminator to make better decisions and reduced the chance of mode collapse. Notably, they were using un-paired data, but it seems that location specific detail is necessary for the CycleGAN to perform proper intensity projection between the two imaging modalities. It remains unclear what benefit this approach provides over the case when the cross-modality images are first registered, i.e., enforcing structural consistency when inputting the data to the network, and then applied within the conventional CycleGAN.

Pan et al. [72] considered more specifically landmarks in images, to aid the estimation of PET from MRI using registered datasets with a 3D patch-based network involving cycle-consistency loss. They proposed Landmark-based Multi-modal Multi-Instance Learning (LM3IL) network to learn the features difference from both MRI and PET for Alzheimer Disease diagnosis. They evaluated PSNR and classification accuracy based on the real and estimated PET images. The authors suggested the method to be sufficiently reliable for clinical use. A comparison of estimated and real images is provided in Fig. 7.

Choi and Lee [74] adapted the pix2pix GAN to generate missing MRI from the florbetapir PET tracer images for Alzheimer's disease patient data. Despite a high SSIM between actual and the estimated MRI, images were blurry and lacked the contrast observed in real MRI. This may be attributed to the lack of detail in the PET, as the MRI had higher resolution. Nonetheless, the approach still had utility for amyloid quantification. In a multiple sclerosis study on myelination, Wei et al. [73] estimated PIB-PET data using a combination of two 3D cGANs called the Sketcher-Refiner GAN. In this network, the Sketcher generates a preliminary image capturing anatomical and physiological aspects of the brain, whilst the Refiner enhances the preliminary image to better reflect myelin content. The choice of the proposed network was to overcome the challenge of predicting myelin changes in PIB-PET from structural MRI due to poor signal and a weak anatomical relationship between PIB-PET and MRI/CT. They also added the L1-based cross entropy loss function. The method outperformed the CNN proposed by Li et al. [40] as well as a single cGAN without a Refiner network. Nie et al. [44] estimated brain CT from MRI using 3D patch-based FCN followed by adversarial training in the auto-context model. The 3D patch reduced across slice inconsistencies, and structural similarity was enforced by using pre-processed images involving registration. Moreover, their

**Table 5**

Summary of cross-modality image estimation using GANs.

| Author-year | Image estimation    | Architecture         | Remarks                         | Paired cross-modal data (Y/N) | Error (HU)<br>Mean $\pm$ standard deviation                                                                                                                     |
|-------------|---------------------|----------------------|---------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [17]        | MR: CT              | CycleGAN             | Brain                           | N                             | MAE 73.7 $\pm$ 2.3                                                                                                                                              |
| [43]        | CT: MR              | CycleGAN             | Heart                           | N                             | Average dice score for real and synthetic data 0.704                                                                                                            |
| [44]        | MR: CT              | Cascade GAN          | Brain & Pelvic                  | Y                             | Brain: Mean MAE 92.5 $\pm$ 13.9<br>Mean PSNR 27.6 $\pm$ 1.3<br>Pelvic: Mean MAE 39.0 $\pm$ 4.6<br>Mean PSNR 34.1 $\pm$ 1.0                                      |
| [45]        | MR: CT              | cGAN                 | Brain                           | Y                             | Mean PSNR 26.6 $\pm$ 1.2, mean SSIM 0.83 $\pm$ 0.03<br>MAE 89.3 $\pm$ 10.3 (Across the entire FOV)                                                              |
| [46]        | MR: CT              | cGAN                 | Pelvic                          | Y                             | MAE 61 $\pm$ 9<br>ME 2 $\pm$ 8                                                                                                                                  |
| [47]        | MR: CT              | CycleGAN             | Musculoskeletal                 | N                             | MAE 29.781 $\pm$ 1.777                                                                                                                                          |
| [48]        | MR: CT              | CycleGAN (EssNet)    | Abdomen                         | N                             | Median DSC 0.9188                                                                                                                                               |
| [49]        | MR: CT              | CycleGAN             | Brain                           | N                             | Not stated                                                                                                                                                      |
| [50]        | MR $\Rightarrow$ CT | CycleGAN*            | Heart                           | N                             | Dice score estimated CT 74.4%<br>Estimated MR 73.2%                                                                                                             |
| [51]        | MR: CT              | CycleGAN             | Brain & Pelvic                  | Y                             | Brain images: MAE 55.7, PSNR 26.6<br>Pelvic images: MAE 50.8, PSNR 24.5                                                                                         |
| [2]         | CT: MR              | CycleGAN             | Brain                           | N                             | MAE 19.36 $\pm$ 2.73, mean PSNR 65.35 $\pm$ 0.86,<br>mean SSIM 0.25 $\pm$ 0.02                                                                                  |
| [20]        | CT: MR              | CycleGAN<br>UNIT     | Lung                            | N                             | DSC 0.75 $\pm$ 0.12<br>DSC 0.63 $\pm$ 0.22                                                                                                                      |
| [52]        | CT: MR              | cGAN (Pix2pix)       | Brain                           | Y                             | Dice 0.54 $\pm$ 0.23<br>Hausdorff Distance 27.88 $\pm$ 21.00                                                                                                    |
| [53]        | MR: CT<br>CT: MR    | CycleGAN(SynSeg-Net) | Abdomen<br>Brain                | N                             | Median DSC 0.919<br>Median DSC 0.966                                                                                                                            |
| [54]        | MR: CT              | cGAN<br>CAE-GAN      | Head-neck-spine                 | Y<br>N                        | MAE: 69.67 $\pm$ 9.27<br>MAE: 100.62 $\pm$ 7.39                                                                                                                 |
| [3]         | MR: CT              | CAE-GAN (VAE + GAN)  | Brain                           | Y                             | MAE: 81.25                                                                                                                                                      |
| [55]        | MR: CT              | cGAN (Pix2pixHD)     | Pelvic                          | Y                             | MAE 48.5 $\pm$ 6                                                                                                                                                |
| [56]        | MR: CT              | cGAN (MCMP-GAN)      | Head (nasopharyngeal carcinoma) | Y                             | MAE 75.7 $\pm$ 14.6                                                                                                                                             |
| [57]        | MR: CT              | cGAN (Pix2pix)       | Pelvic                          | Y                             | MAE 41.2 $\pm$ 3.7 (0.35T Siemens Magnetom C1)<br>MAE 52.0 $\pm$ 5.5 (1.5T Siemens)<br>MAE 43.7 $\pm$ 6.2 (3T GE Discovery)<br>MAE 48.2 $\pm$ 4.9 (3T GE Signa) |
| [58]        | MR: CT              | cGAN                 | Brain                           | Y                             | MAE 47.2 $\pm$ 11.0                                                                                                                                             |
| [59]        | MR: CT              | cGAN (pix2pix)       | Brain                           | Y                             | MAE 60.52 $\pm$ 13.32                                                                                                                                           |
| [60,61]     | MR: CT              | CycleGAN             | Head and neck                   | Y                             | MAE LECT 79.98 $\pm$ 18.11<br>MAE HECT 80.15 $\pm$ 16.27                                                                                                        |
| [60,61]     | MR: CT              | Multi-Cycle GAN      | Head and neck                   | Y                             | MAE 0.0416<br>ME 0.0340<br>PSNR 39.1053                                                                                                                         |
| [62]        | MR: CT              | cGAN                 | Brain                           | Y                             | MAE 0.030 $\pm$ 0.017<br>MSE 0.010 $\pm$ 0.011<br>PSNR 21.422 $\pm$ 3.964                                                                                       |
| [63]        | MR: CT              | MD-CyleGAN           | Pelvic                          | Y                             | RMSE 47.83 $\pm$ 2.92                                                                                                                                           |
| [64]        | MR: CT              | Pix2pix (SA_GAN)     | Pelvic                          | Y                             | MAE 46.79 $\pm$ 2.76<br>MAE 38.5 $\pm$ 4.9<br>PSNR 30.5 $\pm$ 1.3<br>SSIM 0.90 $\pm$ 0.04                                                                       |
| [65]        | MR: CT              | CycleGAN             | Cardiac                         | N                             | Not stated                                                                                                                                                      |
| [66]        | MR: CT              | ResViT GAN           | Brain & Pelvic                  | Y                             | Pelvic images:<br>PSNR 28.45 $\pm$ 1.35<br>SSIM 0.931 $\pm$ 0.009                                                                                               |
| [67]        | MR: CT              | Pix2pix              | Head and neck                   | Y                             | MAE 26.44<br>PSNR 36.97<br>MSSIM 0.93                                                                                                                           |
| [68]        | MR $\Rightarrow$ CT | CycleGAN             | Head and neck                   | Paired + unpaired             | PSNR = 27.599<br>SSIM = 0.595                                                                                                                                   |
| [69]        | MR $\Rightarrow$ CT | Pix2pix              | Pelvic                          | Paired + unpaired             | MR:CT<br>DSC 0.88 $\pm$ 0.14<br>CT:MR<br>DSC 0.83 $\pm$ 0.15                                                                                                    |
| [70]        | MR: CT              | MR: CT               | Brain                           | Y                             | MAE 65.3 $\pm$ 13.9<br>PSNR 28.5 $\pm$ 2.2<br>SSIM 0.90 $\pm$ 0.03                                                                                              |
| [71]        | MR: CT              | CycleGAN             | Brain                           | N                             | PSNR = 32.335870<br>SSIM = 0.95191956                                                                                                                           |

(continued on next page)

Table 5 (continued)

| Author-year                                         | Image estimation                                | Architecture                 | Remarks    | Paired cross-modal data (Y/N) | Error (HU)<br>Mean $\pm$ standard deviation                                |
|-----------------------------------------------------|-------------------------------------------------|------------------------------|------------|-------------------------------|----------------------------------------------------------------------------|
| [72]                                                | MR: PET                                         | CycleGAN                     | Brain      | Y                             | Mean $\pm$ Std PSNR $24.49 \pm 3.46$                                       |
| [73];<br>2019)                                      | MR: PET                                         | cGAN (Sketcher-Refiner GANs) | Brain      | Y                             | MSE $0.0083 \pm 0.0037$<br>PSNR $30.044 \pm 1.095$                         |
| [74]                                                | PET: MR                                         | cGAN (Pix2pix)               | Brain      | Y                             | SSIM $0.91 \pm 0.04$<br>MAE $0.04 \pm 0.03$                                |
| [75]                                                | MR: PET                                         | GLA-GAN                      | Brain      | Y                             | SSIM $96.88 \pm 0.26$<br>MAE $0.01372 \pm 0.001$<br>PSNR $29.326 \pm 0.06$ |
| [76]                                                | MR: PET                                         | GAN                          | Brain      | Y                             | Not stated                                                                 |
| [77]                                                | (Low resolution PET + MR): super-resolution PET | cGAN (LA-GANs)               | Brain      | Y                             | NC subjects $24.58 \pm 1.78$<br>MCI subjects $25.15 \pm 1.97$              |
| [78]                                                | (Low resolution PET + MR): super-resolution PET | CycleGAN                     | Brain      | N                             | SSIM 0.941<br>PSNR 35.57                                                   |
| <b>Cross-modality image estimation excluding MR</b> |                                                 |                              |            |                               |                                                                            |
| [79]                                                | CT: PET                                         | cGAN (pix2pix-M-GAN)         | Lung       | Y                             | MAE 4.60<br>PSNR: 28.06                                                    |
| [80]                                                | CT: PET                                         | FCN-cGAN                     | Liver      | Y                             | MAE $0.72 \pm 0.35$ , mean PSNR $30.22 \pm 2.42$                           |
| [81,82]                                             | PET: CT                                         | cGAN (MedGAN)                | Brain      | Y                             | MSE 264.6                                                                  |
| [81,82]                                             | PET: CT                                         | cGAN                         | Whole body | Y                             | Mean difference $-1.5 \pm 47.3$                                            |
| [83]                                                | CT: US                                          | CycleGAN                     | Abdomen    | N                             | Not Stated                                                                 |

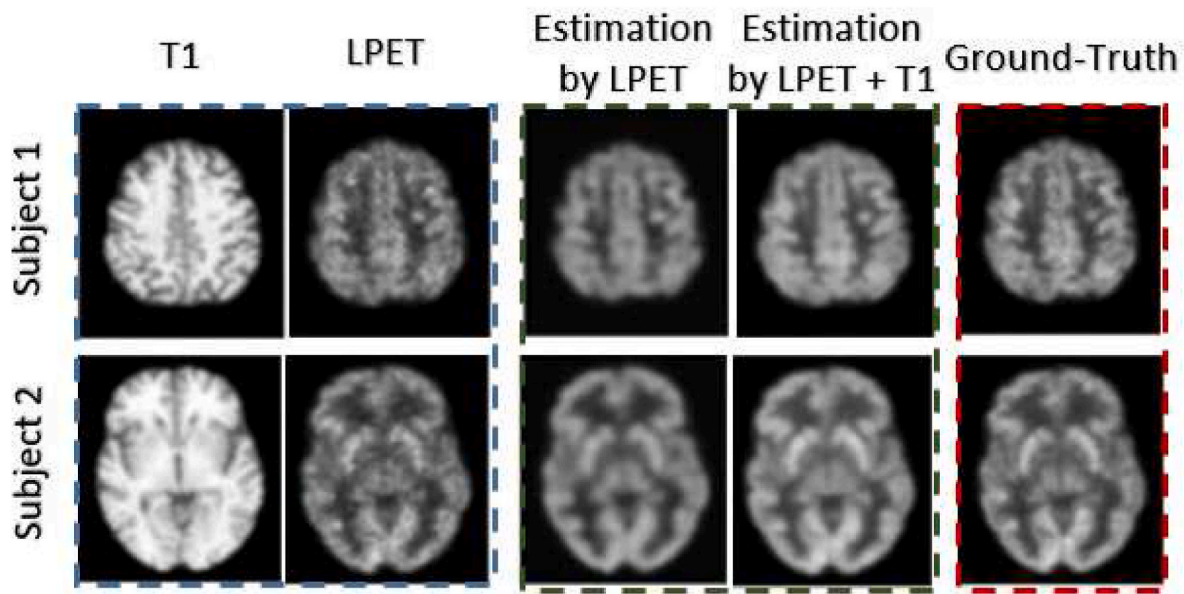


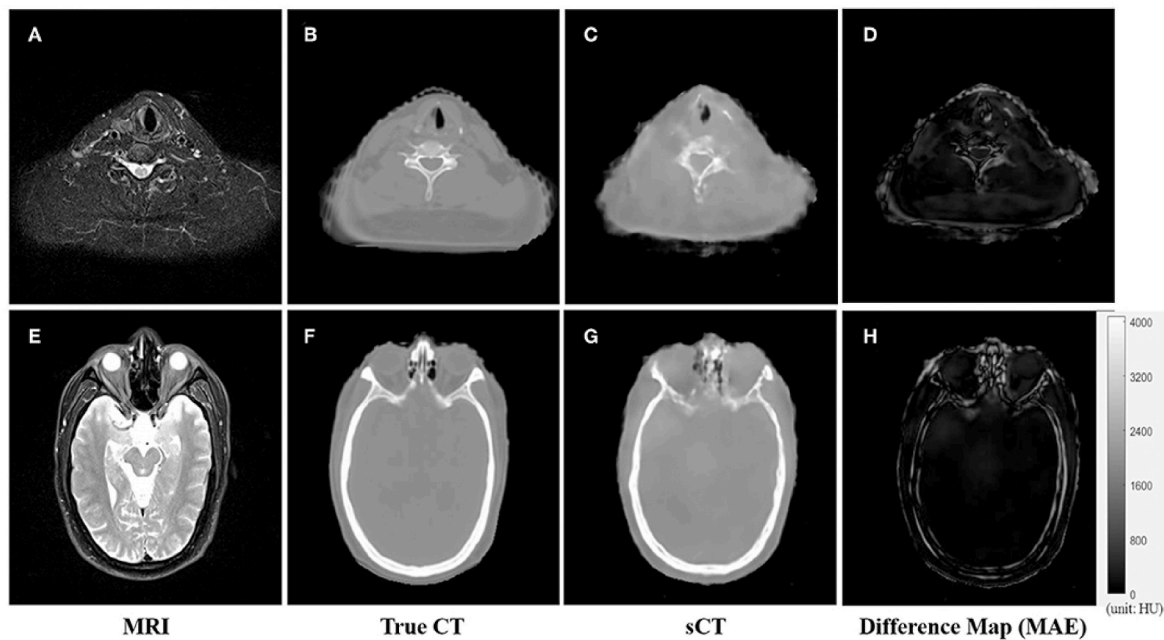
Fig. 3. High-dose PET estimation for two example subjects. Provided are the comparisons between estimated high-dose PET (Ground-Truth) and low-dose PET (LPET), and when estimated without with the use of MRI data (T1-weighted). Adapted and modified from Ref. [6].

primary contribution was the addition of the image gradient difference loss, which helped reduce the blurriness in estimated images. In the same paper the method was applied to estimate 7T MRI from 3T MRI with success, suggesting the intensity projection framework is generalisable. Emami et al. [45] modified traditional GAN by replacing the binary cross-entropy loss function with the least squares version. Subsequently, Lei et al. [51] introduced the Lp norm (mean P distance loss) to replace with L2-norm and integrate with the gradient difference loss. Wang et al. [77] proposed a locality-adaptive cGAN to replace the multichannel input with fused information. The method was applied to predict high dose PET from low dose PET with the aid of MRI. The proposed cGAN consisted of three networks: the locality adaptive network for adjusting spatial differences between MRI and PET images, and generator and discriminator networks. Essentially, the learning of the intensity projection is facilitated through the enforcing of the structural similarity in the two modalities via structure fusion. Synthesised high dose PET images in Fig. 8 are compared against the low dose,

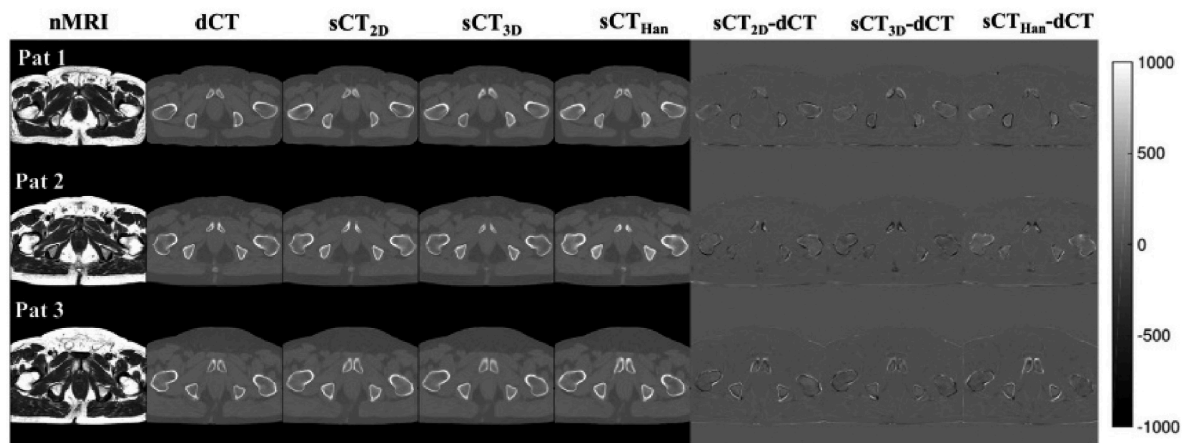
single modality and multichannel PET estimates.

Song et al. [78] implemented a dual CycleGAN (cycle loss and total variation penalty) to improve PET image quality with the aid of MRI. The input was in the form of low dose PET, MRI, high dimensional feature obtained through an auxiliary CNN trained on paired data, and spatial coordinates as landmarks. The network was found to produce high quality images in reduced training times. To overcome the challenge of rigid registration in paired data, and the misalignment issue with unpaired data, Jin et al. [2] developed a dual cycle-consistent and voxel-wise loss CycleGAN using both paired and unpaired data together to estimate MRI from CT. Their result showed improved MAE and PSNR compared with a single CycleGAN using either paired or unpaired data. Subsequently, Sikka et al. [75] showed that image metric improvements can also be achieved through the incorporation of local and global structure similarity constraints within the GAN architecture.

Peng et al. [54] compared the MRI to CT estimation on a nasopharyngeal carcinoma patient cohort using cGAN (paired data) and



**Fig. 4.** An example of real T2-weighted MR and actual (True CT) and estimated CT images (sCT) for the neck (top row) and head (bottom row) regions. The difference map has been provided in the last column [32].



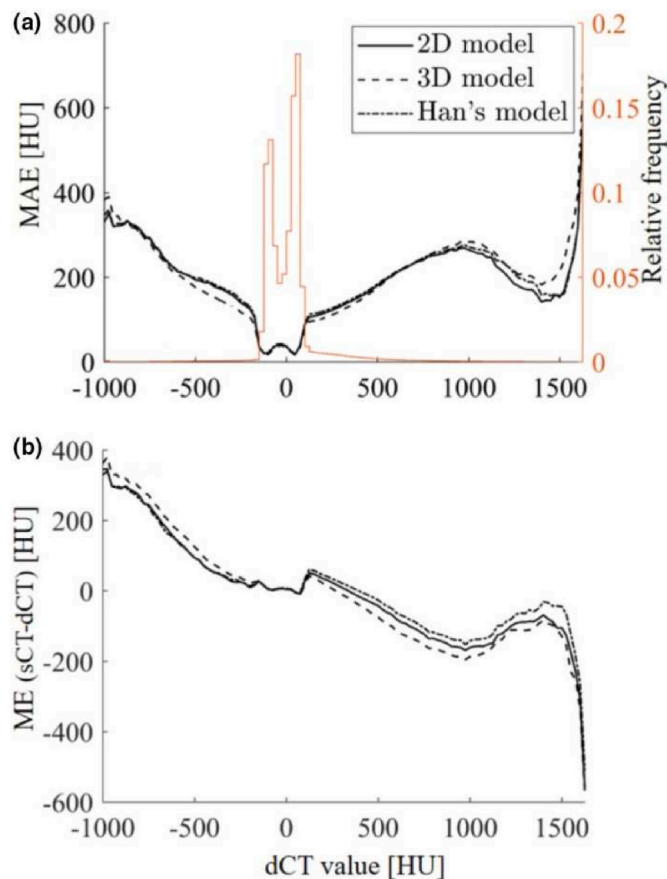
**Fig. 5.** Shown are real T1-weighted MR and real and estimated CT images from 3 patients. From left, normalised MRI images (input); corresponding deformed CT images (ground-truth); estimated CT images (using 2D network); estimated CT images (using 3D network); and estimated CT images (using Han's model). The difference maps between real and estimated CT using each network (i.e., proposed 2D, 3D and Han's) is shown in the last three columns. Adapted from Ref. [30].

CycleGAN (unpaired data). According to the result, they suggested cGAN implementation over CycleGAN when paired high quality cross-modality data is available, see Table 5. Based on these studies, the pairing of the data at the input seems to allow for faster training, and the addition of structural consistency metrics in the form of losses tends to lead to longer training without obvious improvements in image estimation performance. Unpaired input data without spatial information or spatial constraints challenge networks to properly learn the intensity projection between imaging modalities.

CT perfusion and diffusion MRI are used readily in ischemic stroke to map the region affected. Rubin and Abulnaga [52] evaluated how well diffusion MRI can be estimated from CT perfusion scans using a pix2pix FCN-cGAN. They considered two networks one with CT perfusion alone as input and one combined with the estimated MRI for stroke lesion segmentation. The FCN segmentation network with the combined use of actual CT and estimated MRI provided better ischemic stroke lesion outlines than the network with CT perfusion data alone.

Yang et al. [3] combined VAE and GAN and designed a conditional auto-encoder GAN (CAE-GAN) with the view of exploiting the pros of each network. They synthesised CT from multi-contrast MRI data (i.e., fat, water and T2 images). Example image estimates are shown in Fig. 9. The CAE-GAN method resulted in smaller MAE and higher correlation coefficients than fuzzy c-means clustering [88] and the FCN-based deep learning methods [24]. However, spatial qualitative features in images appear similar, and fuzzy c-means clustering produced larger intensities. The primary advantage of using this method appears to be network training based on limited data.

Tie et al. [56] estimated CT from MRI for nasopharyngeal carcinoma patients by implementing 5-level residual U-Net and multi-channel inputs (i.e., pre-contrast T1-weighted, post-contrast T1-weighted with fat-saturation, and T2-weighted MRI). The results of MAE, PSNR and dynamic susceptibility contrast imaging showed outperformance over both multi-channel single-path and single-channel single-path GANs. This result can potentially be attributed to the use of multiple MRI



**Fig. 6.** MAE (a) and (b) mean error of voxels calculated within body masks for estimating sCT from MRI using three models: Fu's 2D model, Fu's 3D model, and Han's 2D model. Reproduced from Ref. [30].

contrasts for CT synthesis, since richer spatial information was utilised to aid the learning of the intensity projection between modalities.

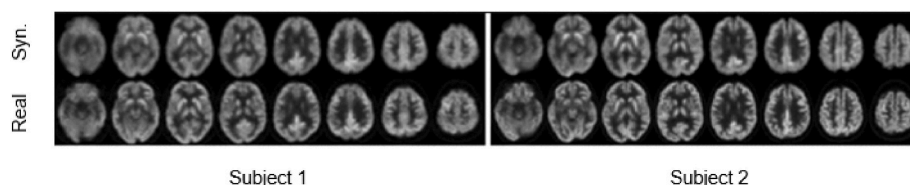
Liu et al. [60,61] addressed the question of estimating, using MRI, the stopping power ratio needed for proton radiation therapy planning. The standard approach is to calculate the ratio from dual energy CT images. They proposed label-GAN consisting of two elements. The elements were namely an MRI to dual energy CT generator (forward

mapping using an FCN-based auto-encoder), a discriminator network (checking low and high energy CTs) and another generator which performed the inverse mapping (dual energy CT to MRI). The first element was constructed as generator-discriminator-generator, starting from MRI, and producing synthetic MRI with the intermediate being dual energy synthetic CT. The second label-GAN element was structured as discriminator-generator-generator using the first network components. Here, the actual dual energy CT was the input, which was past to the discriminator and the inverse mapping generator before it was forward mapped to dual energy synthetic CT. The intermediate image for the second network was synthetic MRI. Prior to training, MRI and dual energy CT were co-registered to achieve spatial consistency between the datasets. The authors recognised that even with co-registration of cross-modality images, spatial mismatch may remain, requiring the use of CycleGAN like losses to highly constrain spatial information between the modalities. The label-GAN was found to produce equivalent level of precision for the stopping power ratio in comparison with the calculation based on actual dual energy CT.

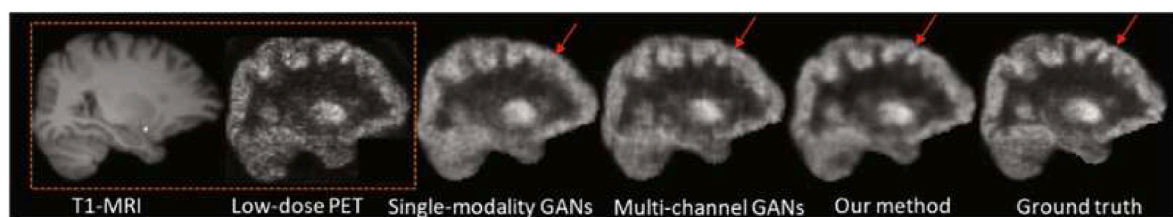
In the application area of CT to PET image estimation, Bi et al. [79] synthesised lung region PET from corresponding CT with a particular focus on creating missing PET slices. They proposed a multi-channel GAN (M-GAN), which takes as input both annotations (labels in the form of high uptake binary mask) and CT of the same slice to synthesise the final PET containing high uptake regions depicting tumour location. We see this method as one of image fusion and synthesis, since PET is synthesised from CT, but the labels must be generated apriori and through a separate process. Improvements in MAE and PSNR in comparison to single channel GAN variants were reported.

Zhang et al. [50] found geometric distortions in image estimation was not mitigated using cycle loss alone. They proposed the additional use of a shape consistency loss obtained from two segmentation networks. Here, each network segments an image modality into meaningful labels, the result of which is implicit shape constraints for anatomical translation. Semantic labels for training images from both modalities were required to train the network. The method was demonstrated to have utility in both MRI to CT, and CT to MRI image synthesis. Primarily, this method is applicable on unpaired data and retains anatomical accuracy. The use of additional data in the form of synthetic images for training led to improved performance.

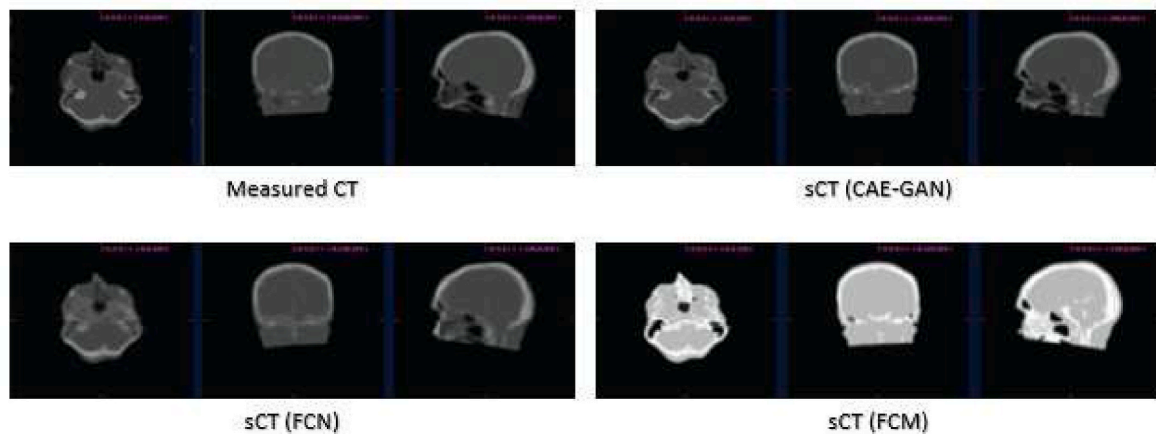
Hiasa et al. [47] considered the synthesis of CT from MRI using a CycleGAN with and without gradient loss and provided qualitative and quantitative results for pelvic region. They found gradient loss to help preserve anatomical shapes in some regions, particularly where high



**Fig. 7.** Comparison of synthetic and real images for two subjects reproduced from Ref. [72].



**Fig. 8.** Comparison of the estimated high dose PET image (Our method is by Wang et al.; low-dose PET fused with MRI) against the acquired Low-dose PET, Single-modality GANs (low-dose PET to high-dose PET training), Multi-modality (low-dose PET and MRI form two channels and target is high-dose PET). Reproduced from Ref. [77].



**Fig. 9.** An example of the CAE-GAN performance compared with FCN and fuzzy c-means clustering (FCM) for synthetic CT (sCT) generation. Images reproduced from Ref. [3].

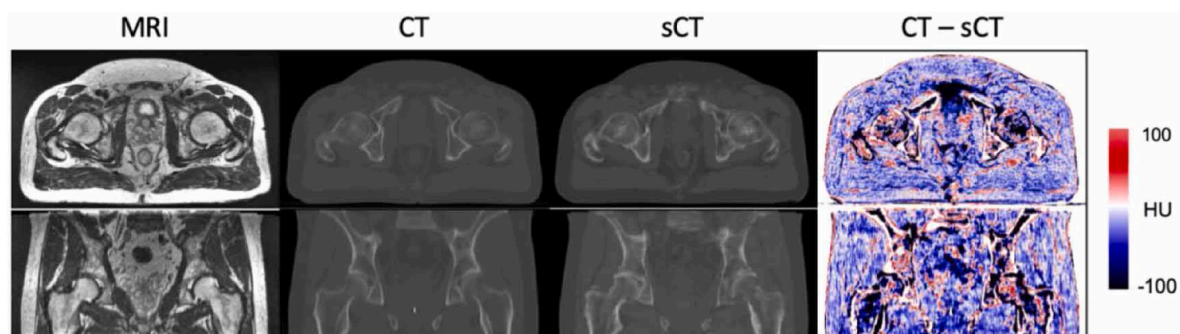
contrast changes were present. They additionally evaluated the use of 20 and over 300 vol for training. The results using 20 vol were already compelling, showing clear anatomical features not too different to those mapped using the CycleGAN trained with 300+ volumes. Whilst the absolute error was shown to reduce with increasing number of datasets, based on their findings it appears plausible that large training datasets may not be necessary to learn intensity projection between imaging modalities.

Brou Boni et al. [55] synthesised the CT image from MRI in the pelvic region using pix2pix cGAN framework (called pix2pixHD) towards establishing an MRI-only workflow for radiotherapy dose calculations. They introduced two auto-encoder generator sub-networks and called it the coarse-to-fine generator, in which one was responsible for translating low-resolution images, and the action of the other was to refine images into a high-resolution output. Notably, this was a study using data from multiple centres, and anatomical mismatches between aligned images were potentially catered for by the coarse representation framework. In addition, a multi-scale discriminator involving three networks was developed in which each function acted at different image scales. For optimisation of the proposed method, the feature matching loss was incorporated into the networks to steady the training by forcing the generator to estimate images with natural statistics at various scales. Finally, cross-entropy cGAN losses in the pix2pixHD network were replaced with Least Square GAN (LSGAN), a key element of which is the optimisation of the gradient vanish during generator updates. Estimated images are shown in Fig. 10. The comparison between estimated and real CT images acquired from patients using pix2pixHD network outperformed Isola et al.'s [15] vanilla pix2pix (MAE  $48.5 \pm 6$  vs  $62.0 \pm 12$  HU; 1.3% dose difference when using synthetic CT versus real CT for

radiotherapy dosimetry).

During the same time, Fetty et al. [57] explored the generalisability of different generators for MRI to CT image estimation for radiotherapy dosimetry, having sourced MRI data from different scanners (Siemens and GE), and additionally, from different field strengths (0.35T, 1.5T and 3T). Data were first rigidly registered to obtain broad alignment, followed by a non-linear registration step. They tested four cGAN generators (i.e., SE-ResNet, DenseNet, U-Net and Embedded Net) and the ensemble model. The ensemble model estimated CT image was of better quality than those produced by any of the other generator networks, with the Embedded Net producing the worst results. MAE results in the body and bone regions confirmed their qualitative observations. Out of the individual generator networks, DenseNet provided the best result followed by the SE-ResNet and U-Net. They noted the importance of evaluating the right metric for the right purpose. They found the MAE metric can discern specific synthesised differences in images, however it was not a predictor of dosimetry differences obtained from the various image estimates. In fact, they drew the conclusion that generator models trained using 0.35T MRI data can be used for dosimetry calculations with 1.5T and 3T MRI data as inputs, irrespective of MRI scanner manufacturer. These findings suggest translatability of MRI to CT synthesis GANs across sites and in time, as scanners are updated with newer software and instrumentation.

Armanious et al. [81,82] noted limitations of previously proposed deep learning models for translating an image to a different modality image. Based on a cGAN they proposed a general image translation tool not specific to imaging modalities (MedGAN). Their inspiration came from other works, having noted that irrespective of the application area, image translation can be achieved using a cGAN provided it is trained

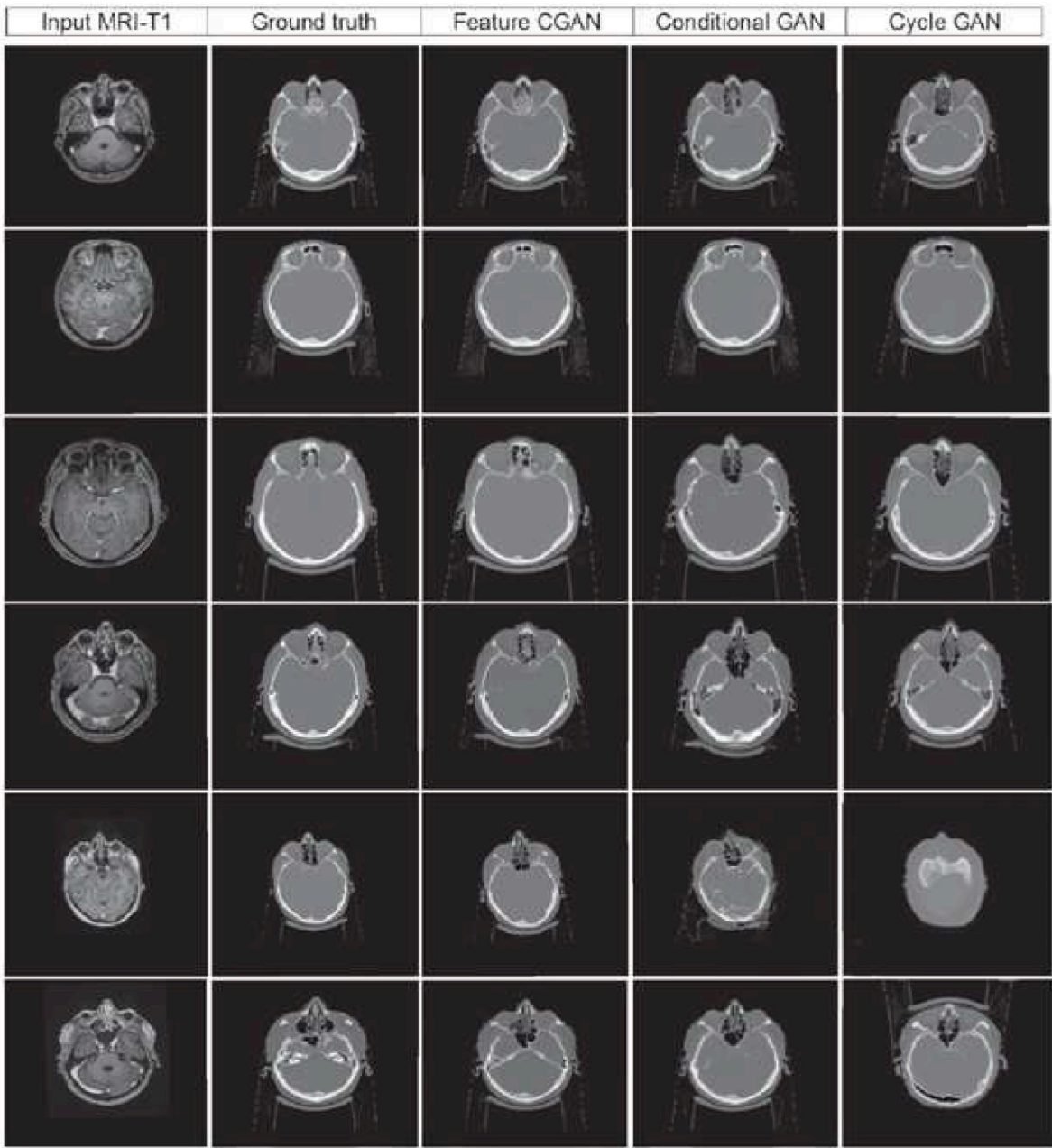


**Fig. 10.** An example of sCT estimated from MRI. The top row and bottom rows represent axial and coronal planes respectively from Ref. [55]. Note, CT values in HU units can range into the thousands.

properly using well-constructed loss functions, and incorporating appropriate generator and discriminator networks. They stated that a cGAN, with a ResNet type generator (CasNet short for cascade Net) is sufficiently general for image-to-image translation. The discriminator was a modified PatchGAN using patches of size  $16 \times 16$ , instead of the previously suggested  $70 \times 70$  patch size. The framework involved a cGAN adversarial loss, supplemented with perceptual, style and content loss. Contributions from style and content losses were weighted orders of magnitude smaller than perceptual loss. The combined use of these losses was suggested to retain image fidelity in view of anatomical structures. The MedGAN image translation framework was successfully demonstrated in PET to CT image synthesis, motion correction in MRI, and denoising of PET images. The current implementation of MedGAN is limited to 2D data, but in the future, it is envisaged it will be extended to work with 3D data as well.

In view of recognising that cross-modality image estimation should

primarily involve the estimation of image intensities for a different imaging modality, researchers have developed methods which aim to constrain learning to the intensity projection alone. In the area of MRI to CT synthesis, and focusing on radiation therapy dosimetry, a highly spatially constrained CycleGAN [60,61], GANs less likely to be sensitive to shape transforms [68], use of different synthetic CT dataset to help the training process converge on the intensity projection [69], local and global discriminator constructs [63], the incorporation of segmentation networks [64], attention mechanisms [70], transformers [66] and neighbourhood descriptors [65] have recently been investigated. These approaches can provide marginal synthetic CT image improvement, and it is difficult to state whether one method is preferred over another. Touati et al. [67] took a different approach, and considered working with intensity histograms. The method is said to provide a spatially invariant approach. In their implementation the machine learning framework seeks to identify the projection between histograms, and as



**Fig. 11.** Comparison of CT estimation from T1-MRI using three methods: the proposed feature cGAN by Touati et al. compared with their implementation of conditional GAN and CycleGAN. Images reproduced from Ref. [67].

such, it is not limited to any specific spatial location. Example images are provided in Fig. 11.

#### 4. Final remarks

Cross-modality image estimation methods have received increasing attention in the field of medical imaging. We reviewed recent advances in the CNN and GAN space. Several things are clear. First, existing methods aim to learn the intensity projection between two imaging modalities. Second, the learning process requires constraints, either through registration, landmarks or novel loss functions, to ensure spatial consistency is retained between the input and output images. Last, GANs appear to have superseded CNNs for cross-modality image estimation.

##### 4.1. General observations and suggestions

Cross-modality images are routinely acquired using different field-of-views and resolutions. This has meant that images have tended to be pre-processed, to match broad alignment and resolution. However, it appears to us that limited pre-processing may suffice. We note that a U-Net, for example, which is an auto-encoder, can compress images to a lower dimension feature space. So, what is important is the feature space, which in theory, can then be expanded into a different resolution image. Using this notion, we argue that resolution matching is probably not necessary between the modalities. This, however, should be tested.

Modern computing hardware used in machine learning can generally handle 2D images. Implementation of networks which take 3D volumetric images is more challenging due to Graphics Processing Unit (GPU) limitations and availability of training data. Three options appear to exist to overcome the challenges. First, divide 3D volumetric images into smaller patches covering the entire 3D volume and create network based on patch data. This reduces the dimension of the image input to the network, the result of which is reduced CNN complexity, at the same time relaxing the computational resource requirements. The number of patches and their size for optimal training of the CNN/GAN remains an open question. With similar benefits, the second approach is to create 2D slices from 3D image volumes, which can later be stacked to recover the 3D image. Lastly, like patches, pooling reduces the number of CNN/GAN parameters, which poses another solution for reduction in computational overheads. A negative consequence of pooling is information loss through a reduction in spatial resolution of image features, an undesirable property for cross-modality image estimation. Nonetheless, the approach may have utility in some cases where high resolution images are intensity projected to inherently lower resolution images (say MRI to PET image synthesis).

It was shown by Isola et al. [15] that patch-based discriminator outperforms a pixelwise synthesis (27% improvement per-pixel accuracy), suggesting that the distributional characteristic of intensities in images are important for cross-modality intensity projection. This observation was later, indirectly confirmed by Touati et al. [67], showing that histogram-based projections are able to produce high quality synthetic images (see Fig. 11). Isola et al. found a patch size of  $70 \times 70$  was able to avoid the tiling artefact. This network was subsequently widely used in other studies (refer to Table 5). It does, however, appear that smaller patch sizes with the use of non-adversarial losses can result not only in sharper images without introducing the tiling artefact. Since different modality images generate different contrast in images, we do not believe these findings are generalisable for all cross-modality image estimations. It may well be that a certain size may sufficiently capture sufficient spatial variation in intensities for robust projection of intensities.

The use of 2D or 3D patches generated from 2D slices or 3D image volumes provides an opportunity to reduce the number of trainable unknowns in the machine learning framework. Nonetheless, an existing shortcoming with patch-based approaches is the lack of a method to reconstitute the 2D slice or the 3D image volume from estimated patch

images (i.e., without the tiling artefact and with broad spatial signal consistency). Moreover, when patches are taken from 2D slices, or in fact the entire slice is synthesised, the current approaches tend to lead to through-slice signal variations due to an intensity offset across estimated slices. Open questions in relation to patches that should be considered in future research include the optimal overlap of patches and how overlaps should be combined, supplementing the work on patch size by Ref. [15]. It seems that an approach which samples patches in a way that each pixel location is covered by the same number of patches has merit. This is because, otherwise, inconsistencies in images may occur due to spatial dependence, for example due to the averaging of different numbers of overlapping patches in different locations. With the use of overlapping patches another problem is likely to arise caused by the number of different image intensities for an adjacent patch based on a defined amount of patch overlap. This may affect the ability to train the machine learning model adequately. The intensity variations for estimated stacked slices requires additional thought. It may be possible to develop approaches involving stacked patches, i.e., across slices, as inputs to the neural network. Otherwise, it may be possible to perform normalization of slices in the estimated images as a post-processing step. Histogram normalization, amongst other methods could potentially be investigated as a solution to this problem. These observations allow us to pose specific questions for the future. Do larger training datasets *per se* mitigate signal intensity variations across patches and slices? Or can smaller training datasets be appended with translations and rotations of patches to create more robustly trained machine learning image synthesis frameworks?

The challenge with intensity projections over different modalities lies in the ability to generate training data which captures disease-related changes. These may, for example, include focal epilepsy, where local changes in MRI and PET signal intensity relate to epileptogenic lesions, oncology where a tumour changes the local image features (both intensity and spatial patterns), and even imaging modality-specific effects such as white matter hyperintensities observed on MRI in multiple sclerosis and vascular dementia. Disorders with abnormal tissue volumes, e.g. brain atrophy in dementia, pose additional hurdles for intensity projection methods. The main challenge in the examples above is obtaining sufficiently large datasets for training and testing. One could therefore, in the spirit of projecting intensity histograms by Touati et al. [67], develop a machine learning framework which does not take images as input and instead estimates the histogram conversion between modalities. The learnt projection profile may then be applied to image patches (2D or 3D). The method may even find utility for entire image slices, and even 3D image volumes, if training data with sufficient histogram resolution can be generated. This may require the crafting of histogram-based training data through MRI to PET simulators, and use clinical data for testing.

##### 4.2. Is cross-modality image estimation valid?

The action of performing intensity projection between images using a neural network is to look for spatial intensity patterns in the image and associate the identified features with patterns expected in the target image. Based on this, image estimation boils down to an intensity pattern recognition and association problem. But one could easily deduce that cross-modality image estimation, in principle, should be the estimation of the correspondence between the physics of different imaging modalities. For example, estimating  $^{18}\text{F}$ FDG-PET from MRI suggests that information across these modalities is consistent. Whilst this may be true for this PET agent in the brain, since tracer uptake localises to all soft tissue classes, and MRI is a soft tissue imaging modality, it may not be the case for a tracer specific only to, say, cancer. Another example is the reduction in  $^{18}\text{F}$ -DOPA uptake in the putamen, a diagnostic marker for Parkinson's disease which is not suitable for MRI to PET image synthesis. The underlying physics of each imaging modality and biological mechanisms driving tracer uptake should always be considered

when developing neural networks for cross-modality image estimation. There has been increasing interest in physics inspired neural networks [89], potentially providing a mechanism by which existing limitations can be overcome. Information from dynamic PET data is often extracted using a kinetic model, which is essentially a set of differential equations describing tracer kinetics in tissue. Hypothetically, one could extend the framework of differential equations to a more general form by incorporating the intensity relationship between the MRI and PET images. In this way, image synthesis could involve a mapping of cross-modality image intensities via an equation learnt from the imaging data.

The validity of the image estimation methods is mostly examined through metrics such as MAE, PSNR and SSIM. In radiotherapy treatment planning, it was shown that these image derived metrics may not reflect an accurate description of how well image estimation performs. It is therefore important to supplement metrics with clinical evaluation. This appears to be routine in image estimations for radiotherapy dosimetry, i.e., estimating CT from MRI, where it was found that image estimates with worse metrics were still useful for dosimetry calculations (<1% accuracy irrespective of the methods considered). It is thus important to involve experts early, namely clinicians and radiologists, to provide a complimentary analysis of findings.

#### 4.3. Which GAN?

The cGAN framework appears to produce better image estimates than conventional GAN, based on both qualitative and quantitative findings. The two primary types of cGANs are pix2pix and CycleGAN. The pix2pix network has been shown to perform better on paired data [90,91], whereas CycleGAN appears to be an ideal choice for unpaired data [17]. The pix2pix architecture has less complexity than CycleGAN, thus requiring less computational resources. The CycleGAN is more complex in structure, suggesting the requirement for larger training datasets.

An important factor is the selection of a generator network with capability of sufficiently preserving structural and spatial information. FCN networks appear to meet the generator requirements [92]. FCN networks using residual connections (i.e., U-NET based) have shown to outperform others (refer to table A adopted from Ref. [15]). In addition, loss functions are also linked to the training and estimation accuracy. We should point out that our review of the existing literature leads us to believe that registration and spatially constrained loss functions achieve a similar outcome, that is, constrain the learning of the intensity projection during training. As such, these two approaches demonstrated on paired and unpaired data may well be equivalent. As such, avoiding of registration prior to network training is likely achievable through an appropriate choice of loss function. Furthermore, it was suggested that cross-modality image estimation is a harder task outside of the brain, where local deformation and variations across datasets can be large.

## Appendices.

### A. Literature search methods

We followed PRISMA-S guidelines [93] for our literature search. PubMed, Scopus, and Google Scholar were searched between 2010 and 2021. Citation searching was also implied to ensure the validity of the literature findings as well as adding articles published in 2022. In the database search, first duplicated studies were removed. Next, remaining articles were screened via title/abstract. From this category, all articles full texts were screened to identify those studies to be included in the review. In citation search, all titles/abstracts were screened first. From the selected list, all articles full texts were also screened, and final list was created to include in the review.

#### A.1 Literature search

To find all the relative studies for our review, we searched based on the following alternative words via UQ library in PubMed, Scopus, and Google Scholar databases.

(structural imag OR molecular imag OR clinical imag OR medical image) AND (image OR imaging) AND (cross-modal OR cross modal OR multimodal OR multi-modal OR intermodality OR intermodal) AND (image estimation OR image prediction OR image generation OR reconstruction

This observation also highlights the importance of spatially constraining the learning of the intensity projection.

We highlighted in this review that 20 image volumes can be used to achieve good image estimates, and by using as much as 300+ image volumes, further, somewhat marginal improvements, can be achieved. Moreover, synthetic images in addition to real images for training can improve cross-modality image estimation performance. Appropriate choices for loss functions in the cGAN framework can provide opportunities to improve the learning of the intensity projection between two medical imaging modalities, especially with limited availability of training data.

#### 4.4. Where to from here?

Future work on image estimation should consider the improvement of the training accuracy by optimising patch characteristics, investigating the impact of the number of datasets used for training, and accelerate training and prediction times by creating more efficient neural networks. People may also wish to consider the application of existing networks to anatomical regions outside of those investigated to date.

Moreover, the true power of cross-modality image estimation may only be realised when pre-processing steps, such as registration and resampling, are no longer required due to adequacy of loss functions capable of sufficiently constraining the learning of the intensity projection between the two different modality medical images.

In summary, cross-modality medical image estimations have widely been investigated using CNNs and GANs. The various studies reviewed used different cross-modality image types, inconsistent training and testing dataset sizes, and images generally consisted of different anatomical locations. Nonetheless, it does appear that cross-modality image estimation has utility in medical imaging, and it should be explored further by targeting specific clinical applications of medical images. Our suggestions for future research should assist the formulation of informative research studies in this area.

## Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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OR translation) AND (pseudo OR synthetic OR synthesis OR artificial) AND (MRI OR CT OR PET OR SPECT) AND (artificial intelligence OR AI OR machine learning OR ML OR deep learning OR convolutional neural network OR neural network OR CNN OR artificial neural network OR ANN OR Adversarial Network OR GAN) NOT radiomics NOT outcome prediction NOT intramodality.

The following conditions were excluded after the search: i) non-medical image estimation, ii) non-CNN and non-GAN based image estimation methods, iii) hybrid PET-MRI scan as an input and other PET (for instance high-dose PET) scan as a ground truth, iv) intramodality studies (for instance estimation of one MRI contrast from another).

## B. Literature search results

The literature search report is provided in Figure 12 based on PRISMA-S [93].

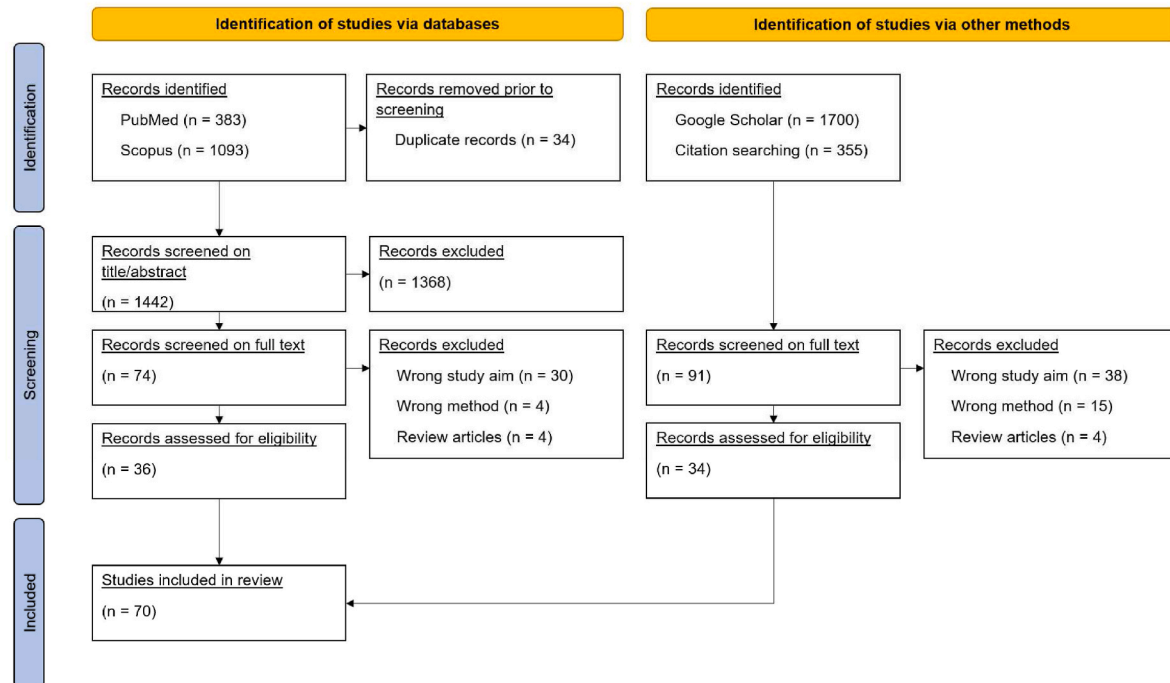


Fig. 12. Literature search diagram based on the PRISMA-S guidelines.

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